ARJUNA JEE 2027

Lecture -05 | Mathematics

BINOMIAL THEOREM





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Topics *to be covered*

A **Properties of Binomial Coefficients**

B **Series**

C **Question Practice**

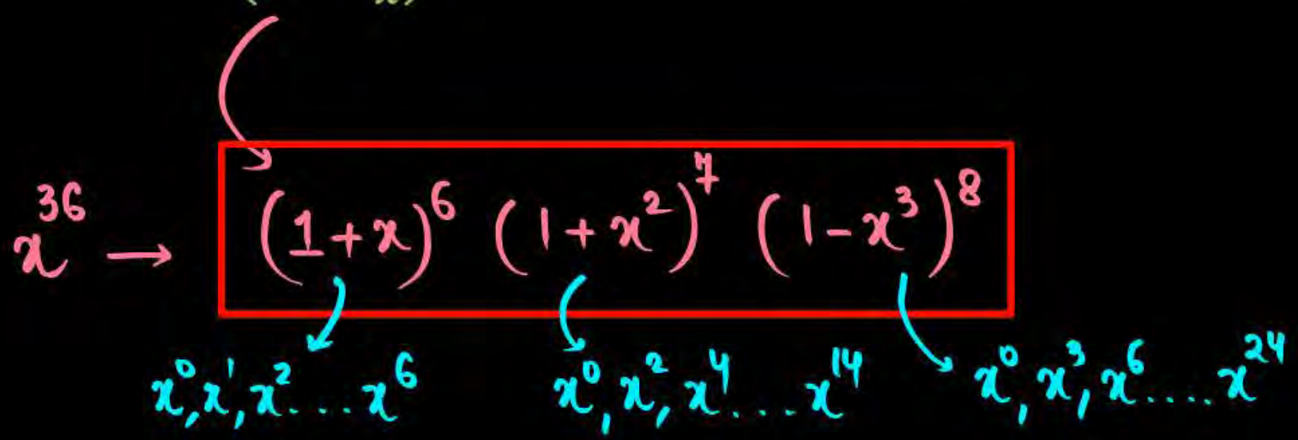
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H.W. Discussion

Question [JEE Mains-2024]

If the coefficient of x^{30} in the expansion of $\left(1+\frac{1}{x}\right)^6\left(1+x^2\right)^7\left(1-x^3\right)^8$; $x \neq 0$ is α, then $|\alpha|$ equals



$x^0, x^1, x^2 \ldots x^6$ $\quad x^0, x^2, x^4 \ldots x^{14}$ $\quad x^0, x^3, x^6 \ldots x^{24}$

$x^{36} \rightarrow (1+x)^6 \, (1+x^2)^7 \, (1-x^3)^8$

x^0	x^{12}	x^{24}
x^2	x^{10}	x^{24}
x^4	x^8	x^{24}
x^6	x^6	x^{24}
x^1	x^{14}	x^{21}
x^3	x^{12}	x^{21}
x^5	x^{10}	x^{21}

$x^4 \quad x^{14} \quad x^{18}$

$x^6 \quad x^{12} \quad x^{18}$

dengluy!!

DIBY-01

In the binomial expansion of $(a-b)^n, n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals

(A) $\frac{6}{n-5}$ (B) $\frac{n-5}{6}$ (C) $\frac{n-4}{5}$ (D) $\frac{5}{n-4}$

[Ans. : (C)]

$T_{r+1} = {}^nC_r\, a^{n-r}(-b)^r$.

$T_5 + T_6 = 0 \Rightarrow {}^nC_4\, a^{n-4}(-b)^4 + {}^nC_5\, a^{n-5}(-b)^5 = 0$

(T_5: $r=4$, T_6: $r=5$)

${}^nC_4\, a^{n-4} b^4 = {}^nC_5\, a^{n-5} b^5$

$\frac{a}{b} = \frac{{}^nC_5}{{}^nC_4}$ ✓

RAPID FIRE DIBY'S

DIBY-04

If a_r denotes the coefficient of x^r in the expansion of $(1-x)^{2n-1}$, then the value of $a_{r-1} + a_{2n-r} =$ is.

(A) -1 (B) 1 (C) $2a_r$ (D) 0

[Ans. : (D)]

$T_{r+1} = {}^{2n-1}C_r (-1)^r \cdot x^r$, a_r

$a_{r-1} + a_{2n-r} = {}^{2n-1}C_{r-1}(-1)^{r-1} + {}^{2n-1}C_{2n-r}(-1)^{2n-r}$

$= {}^{2n-1}C_{r-1}\left\{-(-1)^r + \frac{1}{(-1)^r}\right\} = -\frac{(-1)^{2r}+1}{(-1)^r} = \frac{-1+1}{(-1)^r} = 0$

0

RAPID FIRE DIBY'S

already done!!

DIBY-07

Find the coefficient of x^m in $(1+x)^m + (1+x)^{m+1} + \cdots + (1+x)^n$, where $m \leq n$.

(A) ${}^{n+1}C_{m+1}$ **(B)** ${}^{n}C_{m+1}$ **(C)** ${}^{n+1}C_{m}$ **(D)** ${}^{n}C_{m}$

[Ans. : (A)]

DIBY-08 (Notes)

The sum of the coefficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is –

(A) 2.6^{10} **(B)** 3.6^{10} **(C)** 6^{11} **(D)** None of these

[Ans. : (B)]

$$(2x^2-3x+1)^{11} = a_0 + a_1x^1 + a_2x^2 + \cdots\cdots\cdots + a_{22}x^{22}$$

$$(2x^2-3x+1)^{11} = a_0 + a_1x^1 + a_2x^2 + \ldots\ldots\ldots + a_{22}x^{22}$$

$x=1$

$$(2-3+1)^{11} = a_0 + \cancel{a_1} + a_2 + \cancel{a_3} + \ldots\ldots\ldots + a_{22}$$

$x=-1$

$$(2+3+1)^{11} = a_0 - \cancel{a_1} + a_2 - \cancel{a_3} + \ldots\ldots + a_{22}$$

add

$$0 + 6^{11} = 2a_0 + 2a_2 + 2a_4 + \ldots\ldots + 2a_{22}.$$

$$a_0 + a_2 + a_4 + \ldots\ldots + a_{22} = ?$$

$$\frac{6^{11}}{2} = \frac{\overset{3}{\cancel{6}}\cdot 6^{10}}{\cancel{2}}$$

RAPID FIRE DIBY'S

DIBY-09

The coefficient of x^{49} in the expansion of $(x-1)\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2^2}\right)\ldots\left(x-\frac{1}{2^{49}}\right)$ is equal to –

(A) $-2\left(1-\frac{1}{2^{50}}\right)$

(B) +ve coefficient of x

(C) -ve coefficient of x

(D) $-2\left(1-\frac{1}{2^{49}}\right)$



[Ans. : (A)]

Question [JEE Mains-2022]

Let for the (9th) term in the binomial expansion of $(3 + 6x)^n$, in the increasing powers of $6x$, to be the greatest for $x = \frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k + n_0$ is equal to

$(3+6x)^n$, $T_9 \rightarrow$ greatest

$\hookrightarrow r = 8$

$a = 3$

$b = 6\left(\frac{3}{2}\right) = 9$

$\frac{a}{b} = \frac{3}{9} = \frac{1}{3}$

$-1 \leq 8 \leq \frac{3(n+1)}{4}$

$\frac{3(n+1)}{4} \leq 9$

$3n \leq 33$

$n \leq 11$

$32 \leq 3n+3$

$\frac{29}{3} \leq n$

$9.6 \leq n \leq 11$

$\hookrightarrow n = 10, 11$

$\hookrightarrow n_0$

$(3+6x)^{10} \rightarrow T_{r+1} = {}^{10}C_r \, 3^{10-r} \cdot 6^r x^r$

$\frac{\text{coeff of } x^6}{\text{coeff of } x^3} = \frac{{}^{10}C_6 \, 3^4 \cdot 6^6}{{}^{10}C_3 \, 3^7 \, 6^3} = K = 14$

$14 + 10 = 24$ Ans



LAST CLASS RECAP

Property – 01 :

(i) $^nC_r = {}^nC_{n-r}$

(ii) If $^nC_x = {}^nC_y$ then: $x = y$ OR $x + y = n$

Property – 02 :

Addition of Binomial coefficient

$$^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

Property – 03 : Hockey-Stick Identity

$$^mC_m + {}^{m+1}C_m + {}^{m+2}C_m + \ldots\ldots + {}^{m+k}C_m = {}^{m+k+1}C_{m+1}$$

RAPID FIRE QUESTION

Find : $^{2}C_{0} + {}^{3}C_{1} + {}^{4}C_{2} + {}^{5}C_{3} + \ldots\ldots + {}^{13}C_{11} = {}^{14}C_{3}$

or $^{2}C_{2} + {}^{3}C_{2} + {}^{4}C_{2} + {}^{5}C_{2} + \ldots\ldots + {}^{13}C_{2} = {}^{14}C_{3}$.

$\hookrightarrow {}^{3}C_{3} + {}^{3}C_{2}$

$\hookrightarrow {}^{4}C_{3} + {}^{4}C_{2} = {}^{5}C_{3} + {}^{5}C_{2} = {}^{6}C_{3} \ldots$

PROPERTIES OF BINOMIAL COEFFICIENT

Property – 04 : *Jab r ko numerator (multiplication) se gayab krna ho*

$$\#\quad r \cdot {}^{n}C_{r} = n \cdot {}^{n-1}C_{r-1}$$

$$r \cdot \frac{n!}{r!\,(n-r)!} = \frac{n \cdot (n-1)!}{(r-1)!\,\big((n-1)-(r-1)\big)!}$$

$$r \cdot (r-1)! = r!$$

$$\big((n-1)-(r-1)\big)! = (n-r)!$$

$$r\,{}^{n}C_{r} = n\cdot {}^{n-1}C_{r-1}$$

ex:-

① $7\cdot {}^{13}C_{7} = 13\cdot {}^{12}C_{6}$ ✓

② $23\cdot {}^{58}C_{23} = 58\cdot {}^{57}C_{22}$ ✓

③ $8\cdot {}^{11}C_{3} = 8\cdot {}^{11}C_{8} = 11\cdot {}^{10}C_{7}$ ✓



PROPERTIES OF BINOMIAL COEFFICIENT

Property – 05 : *Jab r ko denominator (division) se gayab krna ho*

$$\frac{{}^{n}C_{r}}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$$

ex :-

$$\frac{{}^{15}C_{7}}{8} = \frac{{}^{16}C_{8}}{16}$$

P-04

$$(r+1)\,{}^{n+1}C_{r+1} = (n+1)\,{}^{n}C_{r}$$

$$\frac{{}^{n+1}C_{r+1}}{n+1} = \frac{{}^{n}C_{r}}{r+1}$$

In any expansion :

$$\text{Sum of all coeff.} = (\text{put variable} = 1)$$

$$\text{Sum of coeff. of even powers} = \frac{(\text{put variable} = 1) + (\text{put variable} = -1)}{2}$$

$$\text{Sum of coeff. of odd powers} = \frac{(\text{put variable} = 1) - (\text{put variable} = -1)}{2}$$

PROPERTIES OF BINOMIAL COEFFICIENT

Property – 06 : *Sum of Binomial Coefficients*

$$(1+x)^n = {}^nC_0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \dots\dots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n$$

$x = 1$

$$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots\dots + {}^nC_n$$

$x = -1$

$$0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots\dots + {}^nC_n (-1)^n$$

$$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots\dots + {}^nC_n \quad \text{—(1)}$$

$$0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots\dots + {}^nC_n(-1)^n \quad \text{—(2)}$$

add

$$\# \quad 2^{n-1} = \{ {}^nC_0 + {}^nC_2 + {}^nC_4 + \dots\dots \} \quad \text{—(3)}$$

$$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots\dots + {}^nC_n$$

$$0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots\dots + {}^nC_n(-1)^n$$

Sub

$$2^{n-1} = \{ {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots\dots \} \quad \text{—(4)}$$

PROPERTIES OF BINOMIAL COEFFICIENT

Property – 06

(i) Sum of all binomial coefficients

$$\sum_{r=0}^{n} {}^nC_r = {}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots + {}^nC_n = 2^n$$

(ii) Sum of all coefficients with alternate signs

$$\sum_{r=0}^{n} (-1)^r \cdot {}^nC_r = {}^nC_0 - {}^nC_1 + {}^nC_2 - \ldots\ldots + (-1)^n \cdot {}^nC_n = 0$$

(iii) Sum of coefficients having ever or odd lower suffix

$$C_0 + C_2 + C_4 + \ldots\ldots = C_1 + C_3 + C_5 + \ldots\ldots = 2^{n-1}$$

RAPID FIRE ROUND

1. $^{14}C_0 + {}^{14}C_1 + {}^{14}C_2 + \ldots\ldots + {}^{14}C_{14} = 2^{14}$.

2. $^{14}C_0 - {}^{14}C_1 + {}^{14}C_2 - {}^{14}C_3 + \ldots\ldots + {}^{14}C_{14} =$ zero.

3. $^{14}C_2 + {}^{14}C_3 + {}^{14}C_4 + \ldots\ldots + {}^{14}C_{13} = 2^{14} - ({}^{14}C_0 + {}^{14}C_1 + {}^{14}C_{14}) = 2^{14} - 16$

4. $^{14}C_0 + {}^{14}C_1 + {}^{14}C_2 + \ldots\ldots + {}^{14}C_6 =$

5. $^{14}C_0 - {}^{14}C_1 + {}^{14}C_2 - {}^{14}C_3 + \ldots\ldots + {}^{14}C_6 =$

6. $^{14}C_0 + {}^{14}C_2 + {}^{14}C_4 + \ldots\ldots + {}^{14}C_{14} = 2^{14-1} = 2^{13}$.

RAPID FIRE ROUND

$2^{14} = {}^{14}C_0 + {}^{14}C_1 + {}^{14}C_2 + \ldots + {}^{14}C_6 + {}^{14}C_7 + {}^{14}C_8 + \ldots + {}^{14}C_{12} + {}^{14}C_{13} + {}^{14}C_{14}$

$2^{14} = 2\left\{{}^{14}C_0 + {}^{14}C_1 + {}^{14}C_2 + \ldots + {}^{14}C_6\right\} + {}^{14}C_7$

4. $\underbrace{{}^{14}C_0 + {}^{14}C_1 + {}^{14}C_2 + \ldots + {}^{14}C_6}_{\text{Half - Bino. Coeff}} = \frac{2^{14} - {}^{14}C_7}{2}$

5. ${}^{14}C_0 - {}^{14}C_1 + {}^{14}C_2 - {}^{14}C_3 + \ldots + {}^{14}C_6 = \frac{1}{2}\,{}^{14}C_7$

${}^{14}C_0 - {}^{14}C_1 + {}^{14}C_2 - {}^{14}C_3 + \ldots + {}^{14}C_6 - {}^{14}C_7 + {}^{14}C_8 - \ldots + {}^{14}C_{12} - {}^{14}C_{13} + {}^{14}C_{14} = 0$

$2\left\{{}^{14}C_0 - {}^{14}C_1 + {}^{14}C_2 - \ldots - {}^{14}C_6\right\} - {}^{14}C_7 = 0$

Question [AIEEE-2007]

The sum of the series $^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots + {}^{20}C_{10}$ is

A $\frac{1}{2}{}^{20}C_{10}$

B 0

C $-{}^{20}C_{10}$

D $^{20}C_{10}$

90% DIBY



RAPID FIRE DIBY'S

DIBY-01

The value of the sum $\sum_{r=0}^{11} {}^{23}C_r$ is :

(A) 2^{23} (B) 2^{22} (C) 2^{11} (D) $2^{22} - 1$

[Ans. : (B)]

DIBY-02

The value of the sum $\sum_{r=0}^{11} {}^{24}C_r$ is :

(A) 2^{23} (B) $2^{23} - {}^{24}C_{12}$ (C) $2^{23} - \frac{1}{2}{}^{24}C_{12}$ (D) $2^{24} - {}^{24}C_{12}$

[Ans. : (C)]

RAPID FIRE DIBY'S

DIBY-03

Evaluate the value of expression : $\left(\frac{C_0+C_1}{C_0}\right)\left(\frac{C_1+C_2}{C_1}\right)\left(\frac{C_2+C_3}{C_2}\right)\ldots\left(\frac{C_{n-1}+C_n}{C_{n-1}}\right)$

(where $C_r = \binom{n}{r} = {}^nC_r$)

(A) $\frac{(n+1)^n}{n!}$ (B) $\frac{n^{n+1}}{(n+1)!}$ (C) $\frac{(n+1)^{n-1}}{n!}$ (D) $\frac{(n+1)^n}{(n-1)!}$

[Ans. : (A)]

PROPERTIES OF BINOMIAL COEFFICIENT

Property – 04 : *Jab r ko numerator (multiplication) se gayab krna ho*

$$\#\quad r \cdot {}^{n}C_{r} = n \cdot {}^{n-1}C_{r-1}$$

$$r \cdot \frac{n!}{r!\,(n-r)!} = \frac{n \cdot (n-1)!}{(r-1)!\,\big((n-1)-(r-1)\big)!}$$

$$r \cdot (r-1)! = r!$$

$$(n-1)-(r-1) = n-r$$

Type-01 : When Binomial Coefficients are multiplied by a constant

Question # $C_r = {}^nC_r$.

$1\,C_1 + 2\,C_2 + 3\,C_3 + 4\,C_4 + \ldots\ldots + n\,C_n =$ _____.

$1.{}^nC_1 + 2.{}^nC_2 + 3.{}^nC_3 + 4.{}^nC_4 + \ldots\ldots + n.{}^nC_n = ??$

$$1 \cdot {}^{n}C_1 + 2 \cdot {}^{n}C_2 + 3 \cdot {}^{n}C_3 + 4 \cdot {}^{n}C_4 + \ldots\ldots + n \cdot {}^{n}C_n = ??$$

$$LHS = \sum_{r=1}^{n} r \cdot {}^{n}C_r = \sum_{r=1}^{n} n \cdot {}^{n-1}C_{r-1} = n \sum_{r=1}^{n} {}^{n-1}C_{r-1} = n \cdot 2^{n-1}$$

$${}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \ldots\ldots + {}^{n-1}C_{n-1}$$

$$1.{}^{n}C_1 + 2.{}^{n}C_2 + 3.{}^{n}C_3 + 4.{}^{n}C_4 + \ldots\ldots + n.{}^{n}C_n = ??$$

#(M-2:

$$(1+x)^n = {}^{n}C_0 + {}^{n}C_1 x^1 + {}^{n}C_2 x^2 + {}^{n}C_3 x^3 + \ldots\ldots + {}^{n}C_n x^n$$

differentiate both sides w.r.to x:

$$n(1+x)^{n-1} = 0 + {}^{n}C_1.(1) + {}^{n}C_2.(2x) + {}^{n}C_3(3x^2) + \ldots\ldots + {}^{n}C_n\, n.x^{n-1}$$

$x=1$

$$n.2^{n-1} = 1.{}^{n}C_1 + 2.{}^{n}C_2 + 3.{}^{n}C_3 + \ldots\ldots + n.{}^{n}C_n$$

$$1.{}^{n}C_1 + 2.{}^{n}C_2 + 3.{}^{n}C_3 + 4.{}^{n}C_4 + \ldots\ldots + n.{}^{n}C_n = ??$$

#M-2:

$$(1+x)^n = {}^{n}C_0 + {}^{n}C_1 x^1 + {}^{n}C_2 x^2 + {}^{n}C_3 x^3 + \ldots\ldots + {}^{n}C_n x^n$$

differentiate both sides w.r.to x:

$$n(1+x)^{n-1} = 0 + {}^{n}C_1.(1) + {}^{n}C_2.(2x) + {}^{n}C_3(3x^2) + \ldots\ldots + {}^{n}C_n\, n.x^{n-1}$$



$$1.2.{}^{n}C_1 + 2.3.{}^{n}C_2 + 3.4.{}^{n}C_3 + 4.5.{}^{n}C_4 + \ldots\ldots + n.(n+1).{}^{n}C_n = ??$$

MI :

$$\sum_{r=1}^{n} r(r+1).{}^{n}C_r = n\sum_{r=1}^{n}(r+1).{}^{n-1}C_{r-1}$$

$$((r-1)+2)\,{}^{n-1}C_{r-1}$$

$$= n\sum_{r=1}^{n}\left((r-1)\,{}^{n-1}C_{r-1} + 2\,{}^{n-1}C_{r-1}\right) = n\sum_{r=1}^{n}(n-1).{}^{n-2}C_{r-2} + n\sum_{r=1}^{n}2\,{}^{n-1}C_{r-1}$$

$$(n-1)\,{}^{n-2}C_{r-2}$$

$$1.2.{}^{n}C_1 + 2.3.{}^{n}C_2 + 3.4.{}^{n}C_3 + 4.5.{}^{n}C_4 + \ldots\ldots + n.(n+1){}^{n}C_n = ??$$

MI:

$$= n(n-1)\sum_{r=2}^{n} {}^{n-2}C_{r-2} + 2n\sum_{r=1}^{n} {}^{n-1}C_{r-1}$$

$$\left({}^{n-2}C_0 + {}^{n-2}C_1 + \ldots\ldots + {}^{n-2}C_{n-2}\right) = 2^{n-2}$$

$${}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \ldots + {}^{n-1}C_{n-1} = 2^{n-1}$$

$$\boxed{\text{Inf} = n(n-1).2^{n-2} + 2n.2^{n-1}}$$

$$1.2.{}^{n}C_1 + 2.3\,{}^{n}C_2 + 3.4.{}^{n}C_3 + 4.5.{}^{n}C_4 + \ldots + n.(n+1)\,{}^{n}C_n = ??$$

#M-2:

$$(1+x)^n = {}^{n}C_0 + {}^{n}C_1 x^1 + {}^{n}C_2 x^2 + {}^{n}C_3 x^3 + \ldots + {}^{n}C_n x^n$$

differentiate both sides w.r.to x:

$$n(1+x)^{n-1} = 0 + {}^{n}C_1.(1) + {}^{n}C_2.(2x) + {}^{n}C_3(3x^2) + \ldots + {}^{n}C_n\, n.x^{n-1}$$

multiply by x^2

$$n(1+x)^{n-1}x^2 = 1\,{}^{n}C_1.x^2 + {}^{n}C_2.(2x^3) + {}^{n}C_3(3x^4) + \ldots + {}^{n}C_n\, n.x^{n+1}$$

differentiate both sides w.r.to x:

$$2x\,n(1+x)^{n-1} + x^2 n(n-1)(1+x)^{n-2} = 1.{}^{n}C_1(2x) + {}^{n}C_2(2.3.x^2) + {}^{n}C_3(3.4x^3) + \ldots + {}^{n}C_n\, n.(n+1)x^n$$

put $x=1$

$$2n.2^{n-1} + n(n-1)2^{n-2} = 1.2.{}^{n}C_1 + 2.3\,{}^{n}C_2 + 3.4.{}^{n}C_3 + 4.5.{}^{n}C_4 + \ldots + n.(n+1)\,{}^{n}C_n$$

Question [JEE Mains-2023]

Suppose $\sum_{r=0}^{2023} r^2 \cdot {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is



Type-02: *When Binomial Coefficients are divided by a constant*

Question

Show that : $\frac{{}^{n}C_0}{1} + \frac{{}^{n}C_1}{2} + \frac{{}^{n}C_2}{3} + \ldots + \frac{{}^{n}C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$ ✓

$$\text{LHS} = \sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1} = \sum_{r=0}^{n} \frac{{}^{n+1}C_{r+1}}{n+1} = \frac{1}{(n+1)} \sum_{r=0}^{n} {}^{n+1}C_{r+1}$$

$$= \frac{2^{n+1}-1}{n+1}$$

Hence.

$$2^{n+1} = {}^{n+1}C_0 + {}^{n+1}C_1 + {}^{n+1}C_2 + \ldots\ldots + {}^{n+1}C_{n+1}$$

$$- {}^{n+1}C_0 \rightarrow 1$$

Question [JEE Mains-2024]

If $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \cdots + \frac{{}^{11}C_9}{10} = \frac{n}{m}$ with $gcd(n, m) = 1$, then $n + m$ is equal to



Question [JEE Mains-2020]

If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+\ldots+x^{2n})(1-x+x^2-x^3+\ldots+x^{2n})$ is 61, then n is equal to _____.



It is never about end result

It's always about Journey

Thank You

#futureIITians